

Ana Sofía M. Uzsoy

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My research interests include large survey science, galaxy evolution, Lyman-Alpha Emitter galaxies, galaxy-environment interactions, and applications of statistical and machine learning for astrophysics.

Education

Harvard University

PhD, Astronomy & Astrophysics (in progress)

Cambridge, MA, USA
September 2022 - May 2027 (exp.)

- Advisor: Doug Finkbeiner

University of Cambridge

M.Phil, Machine Learning and Machine Intelligence

Cambridge, UK
September 2021 - August 2022

- Advisor: Kaisey Mandel
- Thesis: *Scalable Bayesian Inference for Probabilistic Spectrotemporal Models of Type Ia Supernovae*

North Carolina State University (NCSU)

B.S. Computer Science, B.S. Physics

Raleigh, NC, USA
August 2017 - May 2021

- *Summa cum laude*, Dean's List (All Semesters)
- Minors: Mathematics, Oboe Performance

Work Experience

Google

Software Engineering Intern, Quantum AI team

Santa Barbara, CA, USA (remote)
June 2021 - August 2021

- Wrote new methods to improve quantum computing simulators in Cirq (Google's quantum computing library) by modeling noise from quantum hardware.
- Wrote internal software design documents and public documentation for Cirq.

Google

Software Engineering Intern, Tensorflow/Keras/Kaggle teams

Mountain View, CA, USA (remote)
May 2020 - July 2020

- Created natural language processing (NLP) machine learning code examples using TensorFlow 2 for Kaggle, Google's online machine learning and data science competition platform.
- Designed and led launch of new NLP-focused Kaggle competition that remains active.
- Wrote text generation code featured in Google's Data Analyst Professional Certificate Course.

NASA Langley Research Center

Intern

Hampton, VA, USA
June 2019 - August 2019

- Worked on team for SAGE III ISS (Stratospheric Aerosol & Gas Experiment III on the International Space Station), an atmospheric science experimental apparatus aboard the ISS.
- Wrote data analysis pipeline to identify cloud interference in spectroscopic atmosphere measurements using neural networks with PyTorch.

Achievements

- 2026 **Young Alumni "Rising Star" Award**, NCSU Computer Science Department
- 2026 **Astrostatistics Student Paper Competition Finalist**, American Statistical Association
- 2021 **National Science Foundation Graduate Research Fellowship**, awarded to outstanding graduate students in STEM disciplines at US institutions
- 2021 **Churchill Scholarship**, national scholarship providing funding for American students in STEM disciplines to complete an MPhil at the University of Cambridge
- 2021 **Rodney I. McCormick Award**, awarded by NCSU Physics Department to a graduating senior for excellence in undergraduate research
- 2020 **Phi Beta Kappa**, inducted into national honor society for high-achieving undergraduates
- 2020 **First place poster prize**, McCormick Research Symposium, NCSU Physics Department
- 2020 **NCSU Computer Science Department Faculty Senior Scholar**, awarded to a rising senior based on academic excellence, intellectual breadth, and depth of character
- 2019 **Barry M. Goldwater Scholarship**, a national scholarship awarded to promising undergraduates who plan to pursue a STEM research career
- 2017 **Park Scholarship**, a full-ride merit scholarship to NCSU awarded on the basis of outstanding accomplishments and potential in scholarship, leadership, character, and service

Publications (with links)

FIRST- AND SECOND-AUTHORED PAPERS

- Uzsoy, A. S. M.**, & Villar, V. A. (2026). Photometry is all you need: supernova classification as a mixing problem. [\[arxiv preprint\]](#) (submitted to *The Astrophysical Journal Supplement Series*)
- Uzsoy, A. S. M.**, Saydjari, A. K., Dey, A., Raichoor, A., Finkbeiner, D. P., et al. (2026). Bayesian Component Separation for DESI LAE Automated Spectroscopic Redshifts and Photometric Targeting. *The Astrophysical Journal*, 999(2), 241. [\[Journal Article\]](#)
- Uzsoy, A. S. M.**, Dey, A., Raichoor, A., Finkbeiner, D. P., Ramakrishnan, V., et al. (2025). Effect of local environment on Ly α line profile in DESI/ODIN LAEs. [\[arxiv preprint\]](#) (accepted by *The Astrophysical Journal*)
- Uzsoy, A. S. M.**, Lamman, C., & Weber, M. (2025). Manifold Learning for Cosmic Structures. *Machine Learning and the Physical Sciences Workshop @ NeurIPS 2025*. [\[ML workshop paper\]](#)
- de Soto, K., **Uzsoy, A. S.**, & Villar, V. A. (2024). Sharing Space: A Survey-agnostic Variational Autoencoder for Supernova Science. *Machine Learning and the Physical Sciences Workshop @ NeurIPS 2024*. [\[ML workshop paper\]](#)
- Uzsoy, A. S. M.**, Thorp, S., Grayling, M., & Mandel, K. S. (2024). Variational inference for acceleration of SN Ia photometric distance estimation with BayeSN. *Monthly Notices of the Royal Astronomical Society*, 535(3), 2306–2321. [\[Journal Article\]](#)
- Saydjari, A. K., **Uzsoy, A. S. M.**, Zucker, C., Peek, J. E. G., & Finkbeiner, D. P. (2023). Measuring the 8621 Å Diffuse Interstellar Band in Gaia DR3 RVS Spectra: Obtaining a Clean Catalog by Marginalizing over Stellar Types. *The Astrophysical Journal*, 954(2), 141. [\[Journal Article\]](#)
- Uzsoy, A. S. M.**, Zareiesfandabadi, P., Jennings, J., Kemper, A. F., & Elting, M. W. (2021). Automated tracking of *S. pombe* spindle elongation dynamics. *Journal of Microscopy*, 284(1), 83–94. [\[Journal Article\]](#)
- Uzsoy, A. S. M.**, Rogers, L. A., & Price, E. M. (2021). Radius and Mass Distribution of Ultra-short-period Planets. *The Astrophysical Journal*, 919(1), 26. [\[Journal Article\]](#)

CO-AUTHORED PAPERS

- SDSS Collaboration, Aghakhanloo, M., Aguilar, D., Aird, J., & et al [incl. **A. S. M. Uzsoy**]. (2026). The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper. [\[arxiv preprint\]](#) (submitted to *The Astrophysical Journal Supplement Series*)
- Ortiz, A., Ramakrishnan, V., Lee, K.-S., Dey, A., & et al [incl. **A. S. M. Uzsoy**]. (2026). ODIN: Confirmation and 3D Reconstruction of Six Massive Protoclusters at Cosmic Noon. *The Astrophysical Journal*, 1008(1), 85. [\[Journal Article\]](#)
- Grayling, M., Thorp, S., Mandel, K. S., Dhawan, S., **Uzsoy, A. S. M.**, et al. (2024). Scalable hierarchical BayeSN inference: investigating dependence of SN Ia host galaxy dust properties on stellar mass and redshift. *Monthly Notices of the Royal Astronomical Society*, 531(1), 953–976. [\[Journal Article\]](#)

Presentations

TALKS

NSF IAIFI Summer Workshop, MIT; Cambridge, MA	August 11 2026
Joint Statistical Meeting; Boston, MA	August 2 2026
University of Edinburgh; Edinburgh, Scotland	July 16 2026
DESI Collaboration Meeting; Durham, England	July 13 2026
University of Cambridge Astro Data Science Seminar; Cambridge, England	July 9 2026
AstroAI Workshop, Center for Astrophysics Harvard & Smithsonian; Cambridge, MA	June 17 2026
Galaxy Formation Group Meeting, UC Berkeley; Berkeley, CA	March 18 2026
Stanford KIPAC Visitor Tea; Palo Alto, CA	March 13 2026
DESI Lunch, Lawrence Berkeley National Lab; Berkeley, CA	March 11 2026
ITC Luncheon, Center for Astrophysics Harvard & Smithsonian; Cambridge, MA	February 12 2026
DESI Collaboration Meeting; Tucson, AZ	December 11 2025
AstroAI Seminar, Center for Astrophysics Harvard & Smithsonian; Cambridge, MA	November 3 2025
University of Toronto/CITA Cosmology Lunch; Toronto, Canada	October 10 2025
UT Austin Galaxies & Cosmology Seminar; Austin, TX	October 6 2025
Open SkAI Workshop; Chicago, IL	September 5 2025
University of Cambridge Astro Data Science Seminar; Cambridge, England	July 21 2025
AstroAI Seminar, Center for Astrophysics Harvard & Smithsonian; Cambridge, MA	May 19 2025

Universidad de La Serena; La Serena, Chile	April 24 2025
Escape of Lyman Radiation from Galactic Labyrinths; Crete, Greece	April 11 2025
Princeton Astro Machine Learning Journal Club; Princeton, NJ	February 6 2025
Transient Tea, Center for Astrophysics Harvard & Smithsonian; Cambridge, MA	January 24 2025
DESI Lunch, Lawrence Berkeley National Lab; Berkeley, CA	October 9 2024
Stanford KIPAC Visitor Tea; Palo Alto, CA	October 8 2024
Astromerique Seminar, U. of Montreal Ciela Institute; Montreal, Canada (virtual)	October 1 2024
NSF IAIFI Summer Workshop, MIT; Cambridge, MA	August 13 2024
DESI Collaboration Meeting; Marseille, France	July 12 2024
NSF's NOIRLab; Tucson, AZ	May 24 2024
Harvard CHASC Astrostatistics Seminar; Cambridge, MA	February 7 2024
NCSU Physics Department McCormick Symposium; Raleigh, NC (virtual)	May 5 2021
University of Chicago STEM Research Symposium; Chicago, IL	August 1 2018
Leadership Alliance National Symposium; Hartford, CT	July 28 2018

POSTERS

European Astronomical Society Meeting; Lausanne, Switzerland	June 29-July 3, 2026
NeurIPS Machine Learning and the Physical Sciences Workshop; San Diego, CA	December 6 2025
DESI Collaboration Meeting; Berkeley, CA	July 7-11 2025
Rare Gems in Big Data; Tucson, AZ	May 20-24 2024
243rd American Astronomical Society Meeting; New Orleans, LA	January 7-12 2024
NCSU Physics Department McCormick Symposium; Raleigh, NC (virtual)	May 6 2020
235th American Astronomical Society Meeting; Honolulu, HI	January 4-8 2020
2019 American Society for Cell Biology Meeting; Washington, DC	December 7-11 2019
Triangle Cytoskeleton Meeting; Saxapahaw, NC	September 30 2019
NCSU Spring Undergraduate Research Symposium; Raleigh, NC	April 24 2019
NCSU Physics Department McCormick Symposium; Raleigh, NC	April 22 2019
Triangle Cytoskeleton Meeting; Saxapahaw, NC	September 24 2018
Intel International Science & Engineering Fair; Los Angeles, CA	May 14-19 2017

Teaching & Service

Paper Referee

June 2025 - present

- I have served as a reviewer for AI + science workshops at machine learning conferences NeurIPS, ICML and ICLR, and also for DESI Collaboration-Wide Review.

Leadership Team, AstroAI

February 2025 - present

- Serve on committees and help organize talks and workshops for AstroAI, a center dedicated to interdisciplinary research at the intersection of artificial intelligence and astrophysics at the Center for Astrophysics | Harvard & Smithsonian.

Derek Bok Center Teaching Certificate

July 2025

- Received a teaching certificate from the Derek Bok Center for Teaching and Learning at Harvard. Requirements included participating in three pedagogy seminars, receiving teaching feedback via a classroom video consultation, creating an original syllabus, and writing a teaching statement.

Teaching Fellow, Machine Learning for Astrophysics

Spring 2024, 2025

- Guest lectured, held office hours, graded assignments, provided feedback, and supported students in Astro 205, a graduate course on machine learning techniques for astrophysics at Harvard.

Scientific/Local Organizing Committee, AstroAI Workshop

February 2024 - June 2024

- Helped arrange speakers, coordinated volunteers, reviewed abstracts, and chaired sessions for the 2024 AstroAI Workshop at the Center for Astrophysics | Harvard & Smithsonian.

President, NCSU Women in Physics

August 2020-May 2021

- Organized a peer-mentoring program, speaker series and social events to promote a community of women and gender minorities in the NCSU physics department.

NCSU College of Sciences Student Ambassador

August 2018-May 2021

- Represented the NCSU College of Sciences and physics department at university events for current and prospective students.

Skills

	Python, Julia (advanced)
Programming	C, R, Java, MATLAB (intermediate)
	C++, HTML/CSS, Assembly (familiar)
Technology	$\LaTeX$, Git/GitHub, UNIX, Bash, Mathematica
Miscellaneous	Spanish (fluent), Soldering, Circuit wiring, Bacterial culture